

Hammad Arif

Lahore | Islamabad | 0316-7185460 | hammadarif14244@gmail.com |
github.com/hammad2346

Experience

Junior Frontend Developer — Progronics Solutions (Currently Employed)

- Developing and maintaining frontend features using React and Next.js.
- Building responsive UI components with Tailwind CSS and Shadcn.
- Integrating Firebase for authentication and real-time data handling.
- Collaborating with backend and design teams to deliver production-ready features.

Technical Skills

- **Programming Languages:** Java (Intermediate), Python (Intermediate), C++ (Intermediate), JavaScript (Experienced), Typescript (Intermediate)
- **Web Development:** HTML (Experienced), CSS (Experienced), Tailwind (Intermediate), Vue.js (Intermediate), React (Intermediate), Node.js (Intermediate), Next.js (Intermediate), Babylon.js (Intermediate)
- **Databases:** PostgreSQL (Intermediate)
- **Machine Learning & AI:** NumPy (Intermediate), PyTorch (Beginner), LLM APIs (Gemini), Local LLMs (Ollama), LangChain (Intermediate), LangGraph (Intermediate), Redis (Beginner), Docker (Beginner), MCP, n8n
- **Backend Frameworks:** Spring Boot (Beginner), Express.js (Intermediate)
- **Tools & Libraries:** Git (Intermediate), GitHub (Intermediate), Tkinter (Intermediate), p5.js (Intermediate), JWT (Intermediate), REST APIs (Intermediate), Firebase (Intermediate)

Projects

1. Handwritten Digit & Letter Recognition App (Final Year Project)

- Implemented full image preprocessing pipeline including normalization, noise removal, grayscale conversion, resizing, and custom data augmentation.
- Developed two neural networks: NumPy-based from scratch, and an optimized PyTorch version for faster training and GPU usage.
- Built a Tkinter GUI enabling real-time handwritten input and predictions.
- Created a real-time neural network visualization in p5.js showing neuron activations and

input propagation.

2. AP2T (Advanced Physical & Technical Training Platform)

- Built a full-stack web platform in Next.js with role-based access for admins, coaches, parents, players, and front-desk staff.
- Designed PostgreSQL database architecture and implemented backend APIs using raw SQL and server-side business logic.
- Integrated Firebase authentication, file storage, real-time notifications, and kiosk-based attendance workflows.
- Developed scheduling, payment, analytics, and automated email systems to digitize business operations.

3. RAG Chatbot System

- Built a chatbot system capable of answering queries using Retrieval-Augmented Generation (RAG).
- Implemented vector-based retrieval to provide context-aware responses from external knowledge sources.
- Designed multi-step agent workflows using LangGraph for orchestration and reasoning.
- Integrated LangChain to manage retrieval, prompts, and LLM interactions.

4. Customer Management System (Web App)

- Developed the frontend in React with the intention of later converting it into a desktop app using Electron (early experiment included but not finalized).
- Backend built using Spring Boot with a layered architecture: Controller layer (routes), Service layer (logic & calculations), and Repository layer using ORM with entity classes.
- PostgreSQL used as the main database with proper table mappings through JPA entities.
- Designed responsive UI for efficient customer and workshop record management.

5. MCP Mock Data Generation Server (In Progress)

- Backend server that auto-generates PostgreSQL mock data.
- Supports configurable row counts for flexible dataset creation.
- Still actively learning backend patterns and expanding functionality.

Certifications

Google Python Certificate — Google

Education

BS Computer Science — Government College University Faisalabad (GCUF)